



Maintenance Schedule

Diesel Engine

EPA

12V4000T95

12V4000T95L

12V4000T95R

16V4000T95

Application group 4D

MSE50172/06E

Engine model	kW/cyl.	Application group
12V4000T95	155 kW/cyl.	4D, Frac-operation (continuous operation with low load)
12V4000T95L	162 kW/cyl.	4D, Frac-operation (continuous operation with low load)
12V4000T95R	140 kW/cyl.	4D, Frac-operation (continuous operation with low load)
16V4000T95	140 kW/cyl.	4D, Frac-operation (continuous operation with low load)

Table 1: Applicability

Rolls-Royce Solutions refers to Rolls-Royce Solutions GmbH or an affiliated company pursuant to § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture), and Rolls-Royce Solutions Ruhstorf GmbH.

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Definition of Rolls-Royce Solutions

Rolls-Royce Solutions refers to Rolls-Royce Power Systems AG and its subsidiaries Rolls-Royce Solutions GmbH and Rolls-Royce Solutions America Inc.

Maintenance concept

The maintenance schedule is applicable to products certified in accordance with United States Environmental Protection Agency (U.S. EPA) and California Air Resources Board (CARB) emission regulations.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Non-observance of the maintenance instructions can result in violation of the applicable U.S. EPA and CARB emission regulations. Modification or removal of mechanical or electronic components, or the installation of additional components, as well as the execution of calibration processes which may affect the emissions characteristics of the product are prohibited by applicable U.S. EPA and/or CARB emission regulations.

Although Rolls-Royce Solutions' warranty obligations are not dependent upon the use of a specific brand of spare parts, it is recommended that emissions-relevant components be serviced, replaced or repaired using only Rolls-Royce Solutions approved components or their equivalent.

Emissions-relevant components include, but are not limited to, control units, data records, sensors, injectors, exhaust turbochargers, exhaust flaps and all components of the exhaust gas aftertreatment systems.

Use of spare parts that are not of equivalent quality to Rolls-Royce Solutions approved components may impair the effectiveness of the product's emission control systems. The maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

Except for those maintenance steps designated in this Maintenance Schedule as critical emission-related maintenance, an owner or operator of the product is not required to perform the maintenance steps described in this Maintenance Schedule in order to keep the emission-related warranty valid. Rolls-Royce Solutions will not deny an emission-related warranty claim solely on the basis of an owner's or operator's failure to perform recommended additional maintenance or noncritical emission-related maintenance. However, Rolls-Royce Solutions may deny an emission-related warranty claim for the owner's or operator's failure to perform maintenance steps designated as critical emission-related maintenance or if the warranty claim arises out of failures in emission-related components that have been caused by the owner's or operator's improper maintenance or use, by accidents for which Rolls-Royce Solutions has no responsibility, or by acts of God.

If listed in this maintenance schedule, also partially, the following maintenance tasks are emission critical: WM00090, WM00150 and WM00151.

The maintenance system for products from Rolls-Royce Solutions is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals, as well as the required scope of maintenance tasks, are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety.

Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks must be carried out are specified in terms of operating hours and time limits. The specific maintenance task is triggered when either the operating hours or time limit is first reached.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by a Rolls-Royce Solutions' Service Partner. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications.

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Allocation matrix application group 4D

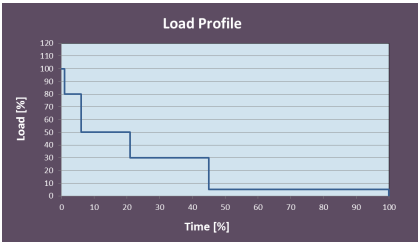
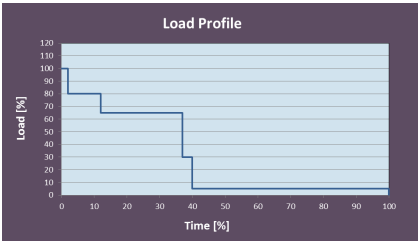
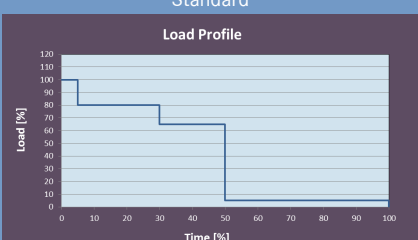
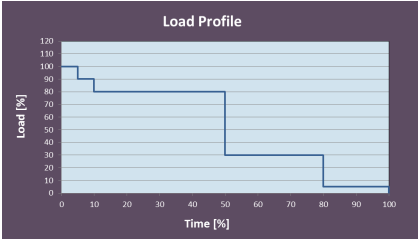
Load profile		12V/16V4000T95 / 12V4000T95x						
Load %	Time %	Load factor %	Load indicator %	121 to 140 kW/cyl.	141 to 163 kW/cyl.			
110 ₍₁₀₀₋₁₁₀₎	—	0 to 22	≤ 2	24,000	18,000	—	—	
100 ₍₉₀₋₁₀₀₎	1							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	5							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	15							
30 ₍₅₋₃₀₎	24							
Idle Speed ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	23 to 30	≤ 4	21,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	2							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	10							
65 ₍₅₀₋₆₅₎	25							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	3							
Idle Speed ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	31 to 41	≤ 9	18,000	15,000	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	25							
65 ₍₅₀₋₆₅₎	20							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	—							
Idle Speed ₍₀₋₅₎	50							
110 ₍₁₀₀₋₁₁₀₎	—	42 to 52	≤ 12	15,000	12,000	—	—	
100 ₍₉₀₋₁₀₀₎	5							
90 ₍₈₀₋₉₀₎	5							
80 ₍₆₅₋₈₀₎	40							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	30							
Idle Speed ₍₀₋₅₎	20							

Table 2: Allocation matrix application group 4D

12V/16V4000T95 / 12V4000T95x							
Load profile		Load factor %	Load indicator %	121 to 140 kW/cyl.	141 to 163 kW/cyl.		
Load %	Time %						
110 ₍₁₀₀₋₁₁₀₎	—	53 to 61	≤ 20	12,000	—	—	—
100 ₍₉₀₋₁₀₀₎	10						
90 ₍₈₀₋₉₀₎	15						
80 ₍₆₅₋₈₀₎	35						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	30						
Idle Speed ₍₀₋₅₎	10						

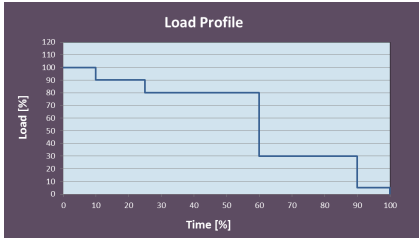


Table 3: Allocation matrix application group 4D

1.4 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0010	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.5 Maintenance Schedule Matrix 24000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00112	1 DAY																														
WM00233	1 DAY																														
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00289	1 DAY																														
WM00290	1 DAY																														
WM00174	1 WK																														
WM00288	1 WK																														
WM00136	1 YR																														
WM00067	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00085	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00170	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X						X						X						X						X
WM00076	3 YR						X						X						X						X						X
WM00664	3 YR						X						X						X						X						X
WM00094	18 YR		X				X						X						X						X						X
WM00776	3 YR								X								X								X						
WM00150	6 YR												X												X						
WM00151	6 YR												X												X						
WM00089	18 YR																X														
WM00090	18 YR																X														
WM00098	18 YR																X														
WM00110	18 YR																X														
WM00093	18 YR																X														
WM00054	18 YR																								X						
WM00056	18 YR																								X						
WM00062	18 YR																								X						
WM00091	18 YR																								X						
WM00096	18 YR																								X						
WM00164	18 YR																								X						
WM00283	18 YR																								X						
WM00921	18 YR																								X						
WM00970	18 YR																								X						
WM00971	18 YR																								X						
WM00973	18 YR																								X						
WM00974	18 YR																								X						
WM00976	18 YR																								X						

DAY = Days

WK = Weeks

YR = Years

HRS = Hours

TIM-ID: 0000000000 - 000